

Artificial intelligence in the workplace – A double-edged sword

Artificial
intelligence in
the workplace

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Abstract

Purpose – The aim of this paper is to outline how artificial intelligence (AI) can augment learning process in the workplace and where there are limitations.

Design/methodology/approach – The paper is a theoretical-based outline with reference to individual and organizational learning theory, which are related to machine learning methods as they are currently in use in the workplace. Based on these theoretical insights, the paper presents a qualitative evaluation of the augmentation potential of AI to assist individual and organizational learning in the workplace.

Findings – The core outcome is that there is an augmentation potential of AI to enhance individual learning and development in the workplace, which however should not be overestimated. AI has a complementarity to individual intelligence, which can lead to an advancement, especially in quality, accuracy and precision. Moreover, AI has a potential to support individual competence development and organizational learning processes. However, a further outcome is that AI in the workplace is a double-edged sword, as it easily shows reinforcement effects in individual and organizational learning, which have a backside of unintended effects.

Research limitations/implications – The conceptual outline makes use of examples for illustrating phenomenon but needs further empirical analysis. The research focus on the meso level of the workplace does not fully refer to macro level outcomes.

Practical implications – The practical implication is that it is a matter of socio-technical job design to integrate AI in the workplace in a valuable manner. There is a need to keep the human-in-the-loop and to complement AI-based learning approaches with non-AI counterparts to reach augmentation.

Originality/value – The paper faces workplace learning from an interdisciplinary perspective and bridges insights from learning theory with methods from the machine learning community. It directs the social science discourse on AI, which is often on macro level to the meso level of the workplace and related issues for job design and therefore provides a complementary perspective.

Keywords Artificial intelligence, Competence, Individual intelligence, Learning theory, Machine learning, Organizational learning, Socio-technical system, Workplace

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1. Introduction

Artificial intelligence (AI) regained high attention during the past years, as it can serve as a technology for not only enhancing learning processes but also changing learning cultures and interactions (Gadanidis, 2017). The cultural component results from the interrelatedness between the concept of knowing and the use of technology. In this regard, AI has the potential to open up for new ways of collaborative knowledge creation (Fischer *et al.*, 2020). The use of AI is obviously more than the integration of a new technological component but unfolds socio-technical system dynamics. There is a need to better understand these dynamics, to explore the collaboration potential but also the risks and unintended consequences that might result from integrating AI in the workplace.

The exploration of socio-technical system dynamics can be related to macro level effects, to workplace issues on meso level and even be explored with a subject-oriented approach on micro level. This paper gives emphasis to the meso level of the workplace. It refers to individual and organizational learning theory to evaluate whether AI can unfold an augmentation potential and support individual abilities instead of substituting or duplicating them. It also searches for the prerequisites to gear into this direction. The core question is how to combine AI and individual intelligence for creating a system with distributed intelligence (Cobb, 1998) and where there are limitations. The course of argumentation is based on



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individual and organizational learning theory as well as competency research which are related to the potential of machine learning methods.

The paper addresses the audience with interest in the use of AI in education. However, the focus is not on dynamics and change issues in educational institutions but on learning and development in the workplace as place and space for competence development. This unit of analysis is thematically related to the overall sociological and economic discourse reflecting on the effects of new technologies on labor and workforce. The research on automation and substitution of human labor due to new technologies and smart machines has a long tradition (Blau *et al.*, 1976; Zuboff, 1988). It has recently been extended to the effects of AI (Ekbja and Nardi, 2017; Susskind, 2020). Generally speaking more pessimistic and more optimistic scenarios coexist in the overall debate (Markoff, 2016; Makridakis, 2017; Wisskirchen *et al.*, 2017). The optimistic view attributes AI to better life (Fischer, 2018), especially if one faces the customer view, e.g. assisted driving systems that enhance drivers and passengers security or AI-based X-ray imaging enhancing the precision of diagnosis for patients. With respect to the quality of work, the authors highlight the augmentation potential of AI for professional work, especially in business, healthcare and education (Markoff, 2016). However, the evaluation of outcomes for the workforce needs a distinct view and has to cope with ambiguities (Wisskirchen *et al.*, 2017) as the more pessimistic scenario gives emphasis to the rationalization of tasks, the risk of a degradation of individual intelligence up to new forms of exploiting labor in computer-mediated networks as networks profit from submitted ideas and data without paying for the input (Perrolle, 1984; Form, 1987; Susskind, 2020; Ekbja and Nardi, 2017).

Even though the evaluation of these effects on macro level is important for the overall debate on new technologies, it does not meet the unit of analysis, as defined in this paper. The focus here is on the learning and development potential provided by the integration of AI in the workplace to the employee or the organization. This also needs a distinct view but referred to criteria of learning and competence development. There are indicators that AI-assisted working systems enhance health and safety, skill development (Wisskirchen *et al.*, 2017) up to new role definition and cross-disciplinary proficiency (Dewey and Wilkens, 2019). But, it is important to note that positive and negative outcomes might differ between stakeholders (Ekbja and Nardi, 2014). As Wilkens *et al.* (2020) show in their empirical exploration of the social acceptance of AI in radiology, there is an overall rather strong belief in the proficiency of AI for better diagnosis and treatment suggestion for the patient among all medical staff members, but only physicians attribute this also to better working conditions while radiographers fear higher techno-stress.

This paper follows a qualitative approach that uses the criteria for evaluating the potential for workplace learning and includes the understanding of individual abilities and technological contributions for augmenting the individual. The key concern of this paper is to figure out whether there is a collaborative learning and development potential while making use of AI in the workplace and where are the challenges and limitations for a valuable use. The focus is on meso level and can therefore complement the ongoing debate on macro level. A core message of the paper is that AI in the workplace tends to function as a double-edged sword, and that it is a matter of socio-technical job design to keep the human-in-the-loop and to complement AI-based learning approaches with non-AI counterparts to reach augmentation.

2. Conceptual outline

2.1 *Why using the metaphor of the double-edged sword?*

There are certain ways for illustrating that something has a light and a dark side at the same time and therefore should be treated with caution. I use the metaphor of the double-edged

sword for describing AI in the workplace because a sword and AI both are man-made tools used by human beings in specific situations to reach targets and perform tasks. The tool is powerful and ambiguous at the same time, and much depends on the intention and way of using it. Double-edged means that there is a high risk of disturbing something unintentionally while using the tool. This is what I want to show in the following discussion. The meaning of a double-edged sword goes beyond a tradeoff, which is a (bargained) outcome while balancing and compromising between different and somehow contradicting goals. I want to underline that there is an entire contradiction in the tool itself. I consciously avoid the metaphor of the “Janus-Faces” (Arnold, 2003) to make clear that AI as understood in this paper is technology and not associated with any kind of personality.

2.2 Understanding artificial intelligence as a tool in the workplace with augmentation potential for human abilities

Human beings always developed and used technology for doing things “better” than without the technical tool. Neanderthal men created first tools from stone and bone for getting food more easily, etc. Even though AI is a much more sophisticated technology, it is a man-made tool developed with an idea of doing something “better” than without this tool. Effects might be different in dependence of the stakeholders in view (Ekbj and Nardi, 2014) and whether one takes a micro-, meso- or macro-level perspective. Outcomes also differ in dependence of the time period that is part of the evaluation. For example, the use of AI as technology can be an issue of efficiency in terms of time and money, which is primarily attributed to employer interests, while negative effects for employees due to the substitution of labor cannot be excluded at least on local level in a short time period. This is currently the case when integrating AI in logistics. Using a long-term perspective and giving emphasis to other stakeholders might nevertheless lead to a completely different estimation for the same practice, e.g. in terms of environmental protection and overall societal welfare. To figure out whether AI has an augmentation potential for human abilities in the workplace, this is not so much an issue of time and money but of criteria such as quality, precision and accuracy, health, safety and motivation, indicating better working conditions (Fischer, 2018; Markoff, 2016). Especially the learning potential provided through AI in the workplace is an important evaluation criterion, which can be directed toward the employee and the organization. Moreover, it is necessary to understand the characteristics and mechanisms of the technology and not to treat it as a black-box to estimate the potential.

AI describes intelligent problem-solving resulting from machine learning, from computer systems building algorithms from big data. According to Hashimoto *et al.* (2018, p. 70), AI “can be loosely defined as the study of algorithms that give machines the ability to reason and perform cognitive functions such as problem solving, object and word recognition, and decision-making.” This also includes self-learning processes on the basis of technical feedback loops and automated interaction with other machines or systems (Wisskirchen *et al.*, 2017).

In some domains, AI occurs more physical and is perceived as hardware while the software component dominates in other domains, even though it is always both. This is why, one distinguishes between physical and virtual AI. Physical AI relates to the function of robots (see Barrett *et al.*, 2012; Thrun, 2004) and the division of labor between machines and human beings. There are industrial robots performing specific tasks primarily in manufacturing, e.g. for metal injection, cutting or pressing. These robots are located in areas separated from human beings. Their use implies a division and sometimes substitution of labor. There is no collaboration with human beings and no learning ability provided. This type of robot is not in the scope of this paper.

Professional service robots go beyond, as they are not separated but closely related to human beings for supporting them, e.g. for precise drug delivery or carrying patients (Hamet

and Trembley, 2017). They can be voice- or eye-activated and could therefore be further developed as personal service robots (Barrett *et al.*, 2012), which nowadays function as AI-based self-learning systems. This type of robot primarily compensates individual disabilities and aims at robot-assisted independent living (Wisskirchen *et al.*, 2017). It has an important support function for human beings and potential for augmentation but not in the meaning of a collaborative work system. The research interest is on the further learning process of the machine from different signals of the surrounding environment while the person depends on this aid. The character of the assistant system is not geared to the learning and development process of the individual in the workplace.

A more recent type is the hybrid robot allowing a close interaction and collaboration with the human being, e.g. in surgeries where the robot assists the surgeon for precision task performance. This is a new form of human-computer collaboration with a mutual development on system level, including the technology, the human being and the overall work system. This type, which is not dedicated to the division of labor but to collaborative job performance, is addressed in the further discussion.

Virtual AI is also dedicated to collaborative task performance. The phrase is typically in use in work processes where it is not so much the physical machine (computer) that is in mind, but the software that is in use, as it is relevant for information processing and decision-making in continuous interrelation with individual work behavior. "The virtual component is represented by Machine Learning, (also called Deep Learning) that is represented by mathematical algorithms that improve learning through experience. There are three types of machine learning algorithms: (1) unsupervised (ability to find patterns), (2) supervised (classification and prediction algorithms based on previous examples), and (3) reinforcement learning (use of sequences of rewards and punishments to form a strategy for operation in a specific problem space)" (Hamet and Tremblay, 2017, p. 37). In further elaboration, Panch *et al.* (2018) show differences in data processing and the level of directed learning when distinguishing between machine learning, deep learning, supervised learning, unsupervised learning and reinforcement learning. The use of virtual AI plays a role in an increasing number of professions. There is especially a rise in fields with high need for information processing such as financial services (Bahrammirzaee, 2010), accuracy in diagnoses and decision-making such as treatment decisions in healthcare (Castaneda *et al.*, 2015; Hashimoto *et al.*, 2018; Topol, 2019) or quality control and business model development in industry (Valter *et al.*, 2018). Different branches profit from similar methods, e.g. AI-based X-ray imaging is equally important for cancer diagnosis in healthcare and quality control of weld seams in industry. The shared point of interest across these use cases is that the overall capacity of AI in information processing, accumulation of experiences and identification of relevant patterns without any signs of fatigue promises to be much higher than that of an intelligent and experienced human being. This is why AI can be considered as a valuable tool for task performance and overall system development augmenting the individual task performance. The major concern of virtual AI is not to bring the human out of the loop but to constitute a human-computer interaction work system of enhanced proficiency with continuous learning and system development (Schuler *et al.*, 2019). There is a need to better understand the collaboration potential between AI and individual intelligence.

2.3 Collaboration potential between virtual artificial intelligence and individual intelligence to be considered in job design

According to Simon's (1996) epistemology, distinguishing between science and science of the artificial AI has to be classified as science of the artificial. Algorithms build on data that represent man-made solutions or outcomes from a man-made social environment. This also includes man-made malpractices and means that algorithms making use of data from a social

context should never be treated as physical law like axioms. However, the background and roots of AI researchers is primarily from science, while individual learning and intelligence as well as individual behavior are described in humanities and social science. This implies that there are completely different perspectives and meanings between these research communities of what learning and intelligence is.

The social science perspective is guiding when considering workplace learning and development. There is a shared understanding that individual intelligence results from cognitive, social and emotional dimensions of learning that define the ability to act in various situations (Boyatzis, 2011; Erpenbeck, 2010). The individual learning process is based on certain activities of information processing, experience, observation, reflection on action and feedback (Bandura, 1989). Knowledge has explicit and implicit components (Nonaka and Takeuchi, 1995), which are embodied and unfold as ability to interact meaningful in a social context (Blackler, 1995). Individuals can combine and transfer their knowledge components to new contexts as a matter of knowledge creation in organizations (Nonaka *et al.*, 2006). Learning processes include system-entire adaptations such as single-loop learning but go beyond in terms of double-loop learning and deuterio learning, which include new target orientations and reflections out of the box (Argyris and Schön, 1978; Argyris, 2003). Individuals underlie bounded rationality due to incomplete and biased information, individual perception in the light of individual norms and beliefs and have an overall bounded capacity of information processing (Simon, 1972). Individuals act according to the perceived environment and former experiences. They are not free from failure, but due to their cognitive, social and emotional ability, have good prerequisites to reflect own and others' experiences, integrate further information and optimize practices while acting and interacting.

Table 1 summarizes the different basic understandings but entire complementarities of AI and individual intelligence.

There are meanwhile scenarios in current AI research to fully duplicate individual intelligence also including emotional components (Goya-Martinez, 2016), but this is not the approach how AI enters the workplace. AI as it is in use in the workplace is based on the research from 20 years ago. This implies that the interest in AI relies on the experiences of the past in a manner of accumulated data. It is domain-specific and not transferrable to any further context (Wilkins Sprafke, 2019). AI is purely based on data and can only be optimized through further data in machine learning and related learning processes. Implicit or tacit knowledge that is not encoded is not included (Vladova *et al.*, 2019). AI needs to be trained for

	Artificial intelligence	Individual intelligence
Basis	Based on big data, availability of data	Cognitive, social and emotional dimension; explicit and implicit knowledge; embodied
Domain	Highly specialized on one context-specific function, no multi-functionality; new combination only as identified context-specific pattern	Flexible, can be transferred to multiple domains as an outcome of knowledge combination and knowledge creation
Learning process	Optimization through data processing in machine learning: deep learning, supervised learning, unsupervised learning and reinforcement learning	Optimization with multiple learning processes and feedback loops, also new goals and system views (single-loop, double-loop and deuterio learning)
Reliability	Rather infinite capacity, no system-specific failure (but based on data from man-made practices of bounded rationality, also accumulation of malpractices of the past)	Bounded capacity, subjective perspective, biased information (risk of failure) but ability to continuously reflect on action and related adaptation

Table 1.
Complementarities
between virtual AI in
the workplace and
individual intelligence

a very specific task or context to provide reliable solutions and, if possible, to communicate its error probability. Under this domain focus, AI has a rather infinite capacity in data processing, which allows to accumulate millions of experiences in comparison to the experience an individual has from own practices or the observation of others. However, as data analytics in fields with man-made solutions is not science but belongs to science of the artificial (Simon, 1996, see above), accumulated data also include wrong decisions and malpractices, which are not always classified as such. This means that AI is not free from failure and wrong decisions, but tends to have a higher capacity to compensate these limitations (Fischer, 2001; Topol, 2019) and at least aims at standards for communicating errors (IEEE Ethically Aligned Design, 2020). Considering the weaknesses of AI, it becomes obvious that it is advantageous to keep the human-in-the-loop. It is only the human being who has the domain-specific, embodied and also tacit knowledge as well as social norms and values as important prerequisites to classify the validity of algorithm-based decision support in the light of former experiences and context factors.

There are complementarities between virtual AI and individual intelligence that help to compensate weaknesses in both directions. Individual intelligence has a weakness in information processing, which can be outperformed by AI. AI misses flexibility between tasks and domains. It is not transferable to any other context, and there is a risk of faulty algorithms and unspecified probability of errors (Topol, 2019; Dewey and Wilkens, 2019). This is the scope for individual intelligence, which shows higher sensitivity within a domain and also allows to cross and transfer between domains to gain new insights.

An example is the use of AI in radiology to classify magnetic resonance imaging (MRI) images. AI can enhance the precision in diagnosis, which is helpful for diagnosing cancer in a very early stage and related treatment suggestions but also for preventing over-therapy in cases of blurry MRI pictures (Dewey and Wilkens, 2019). Even though there is a support function, it is only the medical staff who has further context knowledge of patients and can take into consideration the overall social situation to make a substantial treatment suggestion.

The consequences for integrating AI in the workplace is as follows: it is the accumulation of expertise that makes AI interesting for use in many fields with critical decision-making – such as diagnosis, financial services, insurance, business model development – as long as risk factors resulting from black-box decision are reflected seriously and do not remain underestimated (London, 2019; Toh *et al.*, 2019; Topol, 2019). There is a potential for a fruitful interplay between AI and individual intelligence if AI outbalances the individual weaknesses in information processing and if the individual outbalances the weaknesses of AI in permeating the tacit components of a domain and crossing domain-specific borders. Aiming at better outcome figures such as quality and accuracy depends on the combination of both, AI and individual intelligence.

Making use of the complementarities can especially be fruitful for AI development where there are often experts from the field of machine learning who are not familiar with the domain the algorithms are developed for. As long as the potential of AI is not under- or overestimated, it can be integrated in the workplace as a useful tool for supporting individual task performance, especially if one keeps the human-in-the-loop for the a value-based social reflection of proposed decisions and cross-domain control.

2.4 The contribution of artificial intelligence to enhance individual competencies

Going beyond complementarities between AI and individual intelligence, there is the question whether AI can enhance individual learning opportunities. This makes it necessary to specify the development aims. Workplace learning differs from learning approaches in educational institutions. While the transfer of knowledge in terms of facts, episodes or procedures still dominates in educational institutions (see Fischer *et al.*, 2020), workplace learning aims at the

individual ability to make use of explicit and implicit knowledge components to act efficiently (Erpenbeck, 2010; Wilkens and Sprafke, 2019). The focus is on competencies instead of knowledge and education.

Competency research describes how to classify and operationalize the ability to act when making use of knowledge. These concepts slightly differ in wording but not so much in meaning. For example, Schreyögg and Kliesch (2004) suggest three categories in terms of knowledge interpretation, knowledge combination and cooperation. This concept especially reflects the need to contextualize knowledge. Wilkens and colleagues (Wilkens *et al.*, 2006; Wilkens and Sprafke, 2019) make a distinction between four dimensions: coping with complexity, self-reflection, combination and cooperation. They search for the mechanism of continuously renewing the individual scope of action. In this regard, coping with complexity refers to information processing and target orientation. Self-reflection describes the openness for feedback and a related critical reflection of the state of the art. Combination describes the ability to transfer knowledge to new domains and to combine it for problem-solving activities. Cooperation is the individual ability to extend the individual scope for problem-solving while collaborating with others inside and outside the workplace.

As AI is dedicated to the identification of patterns from big data, it might be supportive for the development of critical competencies. AI tools such as digital assistance systems with automated feedback loops, e.g. in financial services or in complex assembly areas with pick-by-light technology are especially helpful to support information processing and coping with complexity. They can moreover support feedback processes on a technological basis and thus enhance a critical reflection of own behavior or practice in the workplace. AI is not able to transfer knowledge to new domains, and its supportive function for interpretation depends on predefined rules and categories and therefore remains limited. However, AI can support the combination of knowledge, as it can explore failure-sensitive workflow components, which are not explicable for human being. When it comes to cooperation, AI cannot unfold a supportive function, as cooperation is not an issue of nice sounding and steady friendly computer voices. Cooperation is based on social skills and emotional reflection of interaction and a sense for reciprocity, which requires the broader learning scope and embodied knowledge of an individual.

It becomes obvious that AI in the workplace can enhance the development of individual competencies. This is especially the case in information processing, reflection on action and when it comes to the exploration of hidden pitfalls. But, AI has limitations in transferring knowledge to new domains and is unable to contribute to problem-solving via cooperating with others. It tends to be a more valuable tool in situations with high need for vigilance and information processing, e.g. flight control or quality management while working with hazardous substances but has shortcomings in fields with high need for creativity and trust-building communication. This makes the integration of AI in the workplace most challenging, as many workplaces have a high demand in both directions, e.g. hospitals can profit from information processing capacity, but at the same time, might suffer in trust-building communication, as this tends to lack behind if other fields become more prevalent by using AI.

Coming back to the example from radiology with optimized diagnosis of MRI images, AI can train the physicians in correct classification of pictures. But, if the physicians do not understand the process behind and do not reflect the error probability, they cannot use the information for a trustful communication with the patient in a meaningful manner, and they tend to lose their scope for reflecting alternatives in therapy. There is a high demand for integrating the patient in an AI-based decision-making system, which depends on skills in cooperation and interaction. The character of a thoughtful interaction with the patient changes if AI influences the decision-making process and the skills for this new type of cooperation need to be trained on a non-AI basis.

The consequences for integrating AI in the workplace is threefold. First of all, an integration of AI tools can be supportive for individuals who have to cope with high need for information processing, reflection on action and identification of pitfalls or patterns for new solutions. Second, enhancing learning in the workplace cannot be reduced to the use of AI tools. Complementary training approaches are necessary for deeper understanding of the meaning of algorithms ([IEEE Ethically Aligned Design, 2020](#)), but also of out-of-the-box thinking and for team orientation, collaboration and new form of interaction. These latter dimensions of individual competence development are neglected in an AI work environment. Job design has to combine AI and non-AI training approaches. As most workplaces have a demand for all competence dimensions, the complementary training approach seems to be favorable for many fields in industry and services. Third, a pure focus on AI-based learning processes bears a high risk of negative side-effects, as it would enhance some components of individual competencies while neglecting others. This could disturb the balance between the mentioned components of individual competencies that complement each other for enhancing job performance. If the ability to cooperate becomes more and more neglected while finding solutions on the basis of information processing, this would change job characteristics in general and reduce the reliability of work behavior due to a low level of cooperation. The core implication for creating workplace training scenarios is to combine trainings for using AI tools efficiently with other non-AI training elements.

2.5 The contribution of artificial intelligence to organizational learning

AI in the workplace is not just an issue of supporting learning on individual level but also on system level. This system-level learning is the unit of analysis in organizational learning theory. [Levitt and March \(1988, p. 319\)](#) define organizational learning as “encoding inferences from history into routines that guide behavior.” There is core interest in explicit and implicit organizational routines ([Cyert and March, 1963](#)) and in shared mental models of organizational members ([Senge, 2006](#)), which are characterized as organizational knowledge. If there is a transfer from individual behavior to institutionalized organizational practices, there is an underlying organizational learning process. This process takes place if individual experience and practice become an issue of dialogue, language, group norms and shared beliefs of what are good practices within the organization and therefore can guide future behavior. Organizational learning describes the institutionalization from practices to routines that define the organizational knowledge base ([Crossan et al., 1999](#)).

[Argyris and Schön \(1978; Argyris, 2003\)](#) go beyond and make a distinction between different levels of learning and face the change of the organizational knowledge base itself. They describe:

- (1) single-loop learning – this is the adaptation of practices to reach objectives and operationalized targets;
- (2) double-loop learning – describes the redefinition of objectives and targets, including related practices; and
- (3) deutero learning – describes the meta-system of interpreting organizational change processes and their entire logic.

Virtual AI in terms of supervised and unsupervised machine learning has a high potential to explicate the organizational knowledge base as long as it is represented in big data. This is rather the case for explicit than implicit knowledge ([Sanzogni et al., 2017](#)). As AI has a high potential in identifying patterns, there is a support function that can enhance single-loop learning in terms of monitoring practices and outcomes whether they are in line with key targets or not. Current developments in AI even try to encode emotion from voice sounds and

picture elements, including phonation, facial expression or speech usage (Ren, 2009), and try to optimize job design. Methods in reinforcement learning can further enhance this way of single-loop learning. There are especially scenarios in the context of Industry 4.0 that aim at enhancing single-loop learning in the workplace with sophisticated technologically based steering approaches. For instance, a digital assistant system with pick-by-light technology used in complex assembly areas can recognize mental and physical fatigue of the workers and slow down the workflow to prevent from failure. Dopico *et al.* (2016, p. 407) provide a vision of Industry 4.0 with a creation of systems that can perceive their environment and, consequently, can take action toward increasing the chances of success.

Even though a convincing real example is missing for the next point related to deuterio learning, one can argue on a conceptual basis that methods of unsupervised machine learning have a potential in the field of deuterio learning, as they could identify an even unknown pattern of organizational change. Making a hidden pattern visible could support the organization in learning from its own experiences.

With respect to double-loop learning, the shortcomings are obvious: the definition of new targets depends on the managerial decision and is often combined with out-of-the-box thinking. AI-based environmental screenings can prepare double-loop learning but have no initial function integrated in the learning system.

An even more severe limitation is that these three different levels of organizational learning cannot be reached in parallel by using machine learning methods. Training AI in the direction of single-loop learning rather undermines double-loop or deuterio learning. This is the reason why the character of AI as double-edged sword becomes obvious again. Using AI in a specific manner undermines its use in alternative directions. The reinforcement of single-loop learning, which results from the use of AI, lowers the emphasis on double-loop and deuterio learning. This is why, one should treat the promises from self-learning Industry 4.0 factories with caution. If it is just single-loop learning, the overall capacity for system learning will not be only extended but also restricted at the same time.

The implication for integrating AI in the workplace is related to managerial decision-making. AI has a potential for enhancing organizational learning processes, especially for single-loop learning. However, the potential for augmenting organizational learning is limited and unidirectional. Making higher use of AI for single-loop learning undermines alternative approaches and thus defines a need for complementary non-AI approaches to keep managerial thinking dynamic in a non-linear manner.

3. Conclusion

The core outcome from the qualitative evaluation based on the individual and organizational learning theory is that there is an augmentation potential of AI to enhance individual learning and development in the workplace, which, however, should not be overestimated. AI has a complementarity to individual intelligence, which can lead to an advancement, especially in quality, accuracy and precision. Moreover, AI has a potential to support individual competence development as long as AI-based approaches are thoughtfully combined with non-AI approaches. Finally, there is also a potential to enhance organizational learning on a system level with the help of AI. However, a further outcome is that AI in the workplace is a double-edged sword, as it easily shows reinforcement effects that have a backside of rather unintended negative effects. The exploitation of AI for enhancing individual and organizational learning in some fields of development undermines learning and development in other important fields for individual or organizational advancement. This ambiguity of a principally powerful tool implies that the use of AI in the workplace is a challenge for socio-technical job design and managerial decision-making. It depends on how human beings safe human centricity and

keep the human-in-the-loop or the user in control while using AI in specific fields for data analytics. It is also a matter of training design to provide competence development with the help of AI tools, e.g. digital assistant systems, and complement them with non-AI trainings to address a full range of competencies. A deeper understanding of how AI works is important to reflect the power of this tool and to avoid that this power misshapes the socio-technical system, which might happen if the non-AI impulses for new directions and developments are neglected. The implementation and use of AI needs a steering process. An unsupervised implementation process bears not only the risk of negative side-effects but of overall failure due to the failure potential of AI itself. As AI is just a tool and not more, it is the human being who has the freedom and the responsibility to use this double-edged sword in a valuable manner.

4. Limitations and outlook

This paper gives emphasis to the potential of AI for enhancing workplace learning and searches for a collaborative potential between AI and individual intelligence. The outline is based on theoretical insights from the individual and organizational learning theory as well as on conceptual outlines on machine learning. It illustrates theoretically deduced arguments with the help of practical examples, but it does not present empirical data, which allow conclusions with respect to the statistical frequency of AI-augmented work systems. It can be assumed that fields that primarily try to incorporate AI for enhancing individual learning and performance are less frequent than fields that try to benefit from rationalization potential. This has to be explored in further quantitative empirical studies. The core value of this paper is to show that there are also fields for using AI in the workplace that do not primarily aim at rationalization and where are the needs in job design to enhance the augmentation potential. In this regard, the paper prepares ground for operationalizations that do not neglect the augmentation potential.

The paper gives emphasis to the meso level of the workplace and does not fully include the overall macro level discourse on intended and unintended economic and societal effects, as this would be another type of paper. There is no doubt that future research should also evaluate macro level outcomes, in addition to the focus I set in this paper.

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